The following years witnessed the invention of the integrated circuits (ICs) that helped to integrate the entire electronic circuit on a single chip which is small in size and cost-effective. Since 1958 ICs capable of holding several thousand electronic components on a single chip such as small-scale, medium-scale, large-scale, and very-large scale integration started coming into existence. Digital integrated circuits became another robust IC development that enhanced the architecture of computers. All these radical changes led to the introduction of microprocessor in 1969 by Intel.

The electronics revolution, in due course of time, accelerated the computer revolution. Now the world is on its way towards small particles of nano-size, far too small to see. This helps in the miniaturization to an unimaginable size. A room-size computer during its invention has now emerged as a laptop, palmtop, iPad, etc. In the recent past, IBM has released the smallest computer whose size is comparable to the tip of the rice grain, measuring just 0.33 mm on each side.

Electronics is the branch of physics which incorporates technology to design electrical circuits using transistors and microchips. It depicts the behaviour and movement of electrons and holes in a semiconductor, electrons and ions in vacuum or gas. Electronics deals with electrical circuits that involve active components such as transistors, diodes, integrated circuits and sensors, associated with the passive components like resistors, inductors, capacitors and transformers.

This chapter deals with semiconductor devices like *p-n* junction diodes, bipolar junction transistors and logic circuits.

Note

Passive components: components that cannot generate power in a circuit.

Active components: components that can generate power in a circuit.



(a)



(b)



(c)



(d)



(e)

Figure 10.1 Evolution of computers
(a) One of the world's first computers
(b) Desktop computer (c) Laptop computer
(d) Palmtop computer (e) Smallest computer by IBM kept near the tip of the rice grain

DO YOU KNOW

The world's first computer 'ENIAC' was invented by J. Presper Eckert and John Mauchly at the University of Pennsylvania. The construction work started in 1943 and got over in 1946. It occupied an area of around 1800 square feet. It had 18,000 vacuum tubes and it weighed around 50 tons.

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